

How Good Must a Robot’s Model of Your Body Be? Predictive Gentleness Against a Reactive Barrier in Simulated Care Contact

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Abstract—A care robot that touches a person can be gentle in two ways: react to the contact force it measures, or predict how the person’s body will respond before it touches. Reactive control needs no model of the person; predictive control needs one, and the model can be wrong. We ask how wrong it may be. We define gentleness on the recipient’s side, as the involuntary stretch-reflex response of the muscles the contact lengthens, and we measure it in a musculoskeletal simulation of a supported forearm pressed by a robot end-effector. A force-barrier controller protects the force it can observe but not the impact peak, which precedes observation: at any limit it can actually enforce it exceeds that limit in 26 of 28 trials, and at the standard’s own 160 N it never activates at all. A predictive controller that plans against a copy of the recipient’s body reaches the same task at 40% of the barrier’s harm. We then degrade that copy along seven physical axes over a 256-fold range in both directions. Two failure modes appear where preregistered: over-estimating the body makes the robot conservative and the task fails; under-estimating it makes the robot over-confident and harm rises. With reflex prediction alone the over-confident branch crosses the barrier at a four-fold reflex-gain error on one axis and comes within 8% of it at an eight-fold stiffness error in six degrees of freedom. Adding contact-force prediction removes the crossover through a sixteen-fold error on every axis, because contact force depends on the robot’s own stiffness and the contact geometry, which the robot knows. Only muscle strength and passive stiffness need to be accurate; damping, segment mass and reflex delay may be wrong by sixteen-fold without consequence. The result is a specification, not a controller: it says which parameters of the other body a care robot must identify, to what precision, and that it must predict force as well as response.

Index Terms—physical human-robot interaction, assistive robotics, safety, musculoskeletal simulation

A. Introduction

Robots that reposition, wash or dress a person must touch them, and the touch must be gentle. The established route is reactive: a force/torque sensor at the wrist, a limit, and a controller that keeps the measured force under it, today usually a control barrier function (CBF) that renders a safe set forward-invariant [1–3]. Its guarantee has a quiet premise: the harmful quantity is the one being measured. For contact with a limb the peak load is an impact transient that is over before a 50 Hz controller has seen it. A reactive controller is not weak; it is late by construction (Section 4.1).

The alternative is to predict. Ergonomic human-robot collaboration already defines the robot’s objective on the partner’s internal biomechanical state [6–8]; rehabilitation robotics

embeds an OpenSim shoulder in the planner of a robot that moves a passive patient [9], [10]; patient transfer regulates a predicted back-muscle activation to a comfort set-point at 1 Hz [11]. None of these gives the robot a dynamic model of a passive recipient’s *involuntary response to being touched*, uses it to plan the touch before contact, and asks the question that decides whether any of this is deployable: how accurate must that model be before prediction is safer than reaction? The closest studies of model fidelity as a variable find opposite things in other domains: a worse behaviour model made an autonomous car more dangerous [22], while worse motion predictors made a collaborative robot merely conservative, which a distance metric rewards [23].

We answer the question in simulation, with the answer stated as a specification. Contributions:

- 1) **A recipient-side definition of gentleness that a robot can predict.** Gentleness is the rise in activation of the muscles a contact lengthens, relative to the person’s own contact-free baseline, i.e. the stretch reflex the contact provokes. We show why the common co-contraction index cannot serve: a reflex is one-sided and the index only moves when both antagonists rise (Section 3.2).
- 2) **The reactive baseline is late, not weak.** Under a force barrier the steady force respects the limit and the impact peak exceeds it in 26 of 28 trials at 80–120 N, in one and in six degrees of freedom; at the ISO 160 N the barrier activates in 2% of steps and the filtered controller is the unfiltered one (Section 4.1).
- 3) **A fidelity sweep with both failure directions, and where each one bites.** Seven physical axes of the body model, 1/16 to 16-fold, on 19 paired simulated individuals. Under-estimation produces over-confidence and rising harm; over-estimation produces task loss (Section 4.3).
- 4) **Force prediction is what keeps the over-confident branch below the barrier.** An ablation removes it and the branch approaches the barrier at an eight-fold error and crosses it on one axis at four-fold (Section 4.4).
- 5) **A specification.** Only muscle strength and passive stiffness must be identified, to within about four-fold; the robot must predict contact force as well as response; the extra freedom of a six-degree-of-freedom end-effector is a freedom a wrong model misuses systematically (Section 4.5).

B. Related work

Reactive safety. Control barrier functions render a hand-specified safe set forward-invariant and are the standard safety filter for physical human-robot interaction, including versions with an uncertainty margin from a human model (UA-PCBF) and human-aware MPC [1–3]. All of them protect a measured or predicted robot-side quantity: force, distance, energy. We take a force barrier with an online-identified contact stiffness as the baseline and show that the quantity it cannot see, the impact peak, is the one that exceeds the limit (Section 4.1). ISO/TS 15066 supplies the limits; its force column is an AIS-1 injury-onset threshold and its pressure column a pain-onset threshold from a single 100-participant study, and the two must not be described jointly as “pain onset” [4], [5].

Objectives on the partner’s body. Ergonomic HRC defines the robot’s objective on the co-worker’s internal biomechanical state: joint overloading torque predicted from a whole-body model and observed kinematics [6], [7], or muscle activation learned from musculoskeletal simulation [8]. Rehabilitation robotics embeds an analytic OpenSim shoulder with a tendon-strain channel in the online planner of a robot that moves a passive patient [9], [10]; patient transfer regulates a regressor’s estimate of the recipient’s latissimus dorsi activation to a comfort set-point at 1 Hz [11], alongside analytic load models [12]. Robot-side definitions of gentle touch (force ceilings of 3–15 N, C-tactile-optimal stroking at 1–10 cm/s, skin temperature) [13], [14] we keep as constraints, not as the definition.

Reflexes as the recipient’s response. Hodges and Tucker’s reformulation of the motor adaptation to pain, that activity is redistributed rather than uniformly increased, is why we measure the response of the lengthened muscles rather than co-contraction [15]; the co-contraction index correlates with fear, not with pain report [16], and rises with age for reasons of strength rather than threat [17]. Older adults co-contrast more and later under perturbation [18]; complete spinal cord injury amplifies and spatially expands the withdrawal reflex without sensation [19], which bounds the scope of any reflex-based proxy. Stretch-reflex models of the elbow reproduce the resistance of a spastic limb to robot-imposed stretch [20]; an offline musculoskeletal simulation with spindle feedback exists [21] but is not held by a robot.

Model fidelity as a variable. Tian et al. vary the fidelity of a human behaviour model over three levels in simulated driving and find a qualitative crossover against a reachability monitor [22]; Zhao et al. compare four motion predictors against a reactive baseline on real hardware and find no safety crossover, because degraded prediction induces conservatism, which a separation-distance metric rewards [23]. Both directions of failure therefore have to be swept and both task and harm have to be reported, which fixes our design. The only located precedent for varying a human-body model’s fidelity is domain randomisation for exoskeleton control [24]. Caregiving simulators give the simulator a detailed human but the robot no model of it [25–27].

Gap. To our knowledge, no prior work equips a robot with a forward dynamics model of a passive or frail care

recipient’s body that predicts, ahead of contact, the body’s involuntary protective response to robot-initiated contact, uses that prediction for gentle-contact planning, and treats the fidelity of that model as an independent variable. The closest rehabilitation work estimates reflexive reactions after they occur rather than predicting them [10]; the closest transfer work predicts a scalar quasi-static activation without temporal dynamics [11]; the closest reflex-bearing forward model is an offline simulation not held by any robot [21].

C. Method

1) *The recipient:* MyoSuite’s myoArm (38 joints, 63 Hill-type muscle-tendon units, MuJoCo 3.6, 2 ms step). The humerus is welded to the world, the kinematic idealisation of an upper arm resting on a bed; the elbow and hand are free. A stretch reflex on the eight elbow muscles converts muscle-tendon velocity and length excursion, after a 25 ms afferent delay and a 30 ms activation lag, into activation [20]; its gain is fixed once against an independent anchor (a torque increment inside the healthy passive-resistance band) and never retuned. Individuals differ: each seed perturbs muscle strength, passive stiffness, joint damping and segment mass by a log-normal factor of $\sigma = 0.25$. Seeds whose posture leaves no contact-free parking pose are excluded before any arm is run (3 of 22).

2) *Gentleness, measured on the recipient:* The harm proxy L2 is the mean rise in activation, over the muscles whose muscle-tendon length exceeds its pre-contact baseline, relative to that baseline. The co-contraction index 2 min(flexor, extensor) cannot carry this signal: pressing the forearm lengthens one group and shortens the other, so only one responds and the minimum does not move. Measured: welding the humerus raised muscle stretch 1.4-fold and the index’s response fell 8.5-fold while L2 rose. L2 is manner-sensitive where it must be (separation of contact manner from posture noise 943, Spearman +0.70 with approach speed, +0.78 with depth) and reads zero without contact. Peak contact force is reported alongside as the mechanical anchor, with the ISO/TS 15066 forearm limit of 160 N (the AIS-1 injury-onset column, not pain onset).

3) *The robot and the observation boundary:* The end-effector is idealised: a capsule (radius 25 mm, half-length 50 mm) on three slide and three hinge joints under position servos of 6000 N/m and 20 N m/rad, the same translational stiffness as the one-axis probe of the pilot. It parks 40 mm above the forearm, capsule along the bone, and commands a pose every 20 ms. Every controller sees exactly the same observation: its own pose and velocity, the contact force vector (a wrist F/T sensor, 0.5 N noise), and the recipient’s elbow, forearm-rotation and shoulder-elevation angles with 0.01 rad noise. Muscle activation, tendon lengths and the harm proxy are never observable; a counter records every read of ground truth and an audit asserts the count is zero for every controller (a controller that peeks is caught by the self-test).

4) *Task:* Extend the elbow by 15 degrees, about half of what a greedy 1 m/s press to 0.09 m produces. The task is measured on the recipient, so it cannot be gamed by moving

TABLE I
THE REACTIVE BARRIER PROTECTS THE FORCE IT SEES AND MISSES THE PEAK

common limit	constraint arm: peak force	over its own limit	barrier active
80 N	103 N	7/7	9% of steps
100 N	126 N (123 N, n = 19)	7/7	8%
120 N	132 N	6/7	7%
160 N (ISO)	146 N	0/7	2%

the probe along the surface. Task completion is reported as a fraction; harm as the episode peak of L2 and of contact force.

5) Arms:

- **Greedy:** press straight down at 1 m/s until the task is met.
- **Constraint (CBF):** greedy filtered by a force barrier on the measured force vector, with a rate condition and an equilibrium condition using a contact stiffness identified online; when the normal bound binds the excess normal push is removed and the tangential command kept (the QP closed form), so it slides rather than brakes. No model of the recipient.
- **Predictive:** sampling model-predictive control over push direction (down, and down tilted toward either flexion-plane normal and either hinge-axis direction) and rate, scored against a private MuJoCo copy of the recipient that runs alongside the real one and is corrected only by the observations above. Each candidate plan is rolled out in the copy until the copy's elbow meets the task; the plan with most predicted progress wins subject to two predicted budgets, reflex $L2 \leq 0.06$ and contact force $\leq$ the common limit.
- **Fidelity conditions:** the copy's parameters scaled by f on one axis: muscle strength, passive stiffness, joint damping, segment mass, range of motion, reflex gain, reflex delay; f in 1/16, 1/8, 1/4, 1/2, 2, 4, 8, 16; plus the nominal copy ($f = 1$), a copy with the individual's own parameters, and a copy with every parameter randomised (the knowledge-free control).
- **Common force limit:** 100 N for both arms in the main table. At the ISO 160 N the barrier activates in 2% of steps and the constraint arm is the greedy arm; 80, 120 and 160 N are reported as a sensitivity.

All comparisons are paired by seed; 95% intervals are paired-t over 19 individuals (7 in the pilot and ablations noted as such). Directions, criteria, the domain extension rule and every amendment were preregistered (PREREGISTRATION.md).

D. Results

1) *The reactive barrier protects the force it sees and misses the peak:* Steady force respects the limit in every case; the impact transient does not (26 of 28 trials across the four limits below 160 N). Slowing the greedy press ten-fold in the one-axis pilot brings the peak under the limit and drops task completion to 0.67. The overshoot is not a gain-tuning problem: the peak is not observable until it has happened.

2) *Prediction with a good model:* Nineteen paired individuals, common limit 100 N (crossover6_ext_n20.json):

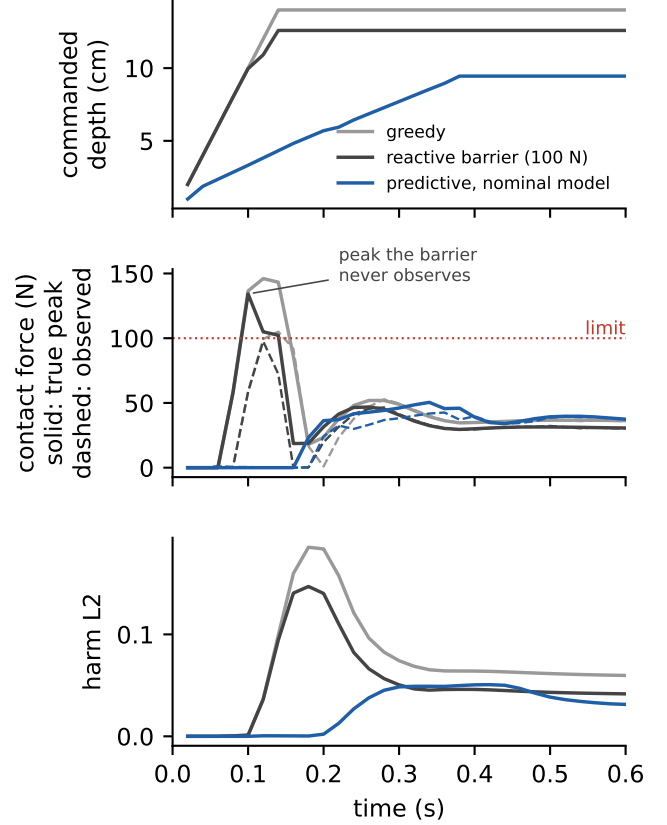


Fig. 4. One individual (seed 0), one episode per arm: commanded depth, contact force (solid: within-step true peak; dashed: what the controller observes at the step boundary) and true L2 against time. The barrier's true peak (134 N) sits above its 100 N limit while its observed force never exceeds 98 N.

TABLE II
PREDICTION WITH A GOOD MODEL

arm	task	harm L2	peak force	paired vs constraint
greedy	1.000	0.168	150 N	-
constraint (CBF)	1.000	0.136	123 N	-
predictive, nominal model	0.967	0.050	50 N	-0.086 [-0.094, -0.078]
predictive, individual's own model	0.998	0.049	48 N	-0.081 (n = 7)
predictive, knowledge-free model	0.30	0.032	29 N	task lost

The predictive arm completes 97% of the task at 37% of the constraint arm's harm and 41% of its force. The nominal body is, for this purpose, the true one: giving the copy the individual's own parameters changes harm by 0.001. The knowledge-free copy predicts the sign of harm reversed (correlation with true harm -0.99 in the pilot) and withdraws.

3) *Both failure directions, where preregistered:* Harm (L2) of the predictive arm against the fidelity factor, 19 individuals, both budgets active. Constraint arm: 0.136.

($f \geq 2$ columns from the 7-seed sweep; 19-seed values for $f \leq 1/2$ and $f = 1$. Full tables in out/audit_numbers.txt.)

Under-estimating the body ($f \downarrow 1$) makes the robot over-confident: harm rises monotonically from 0.050 to 0.092 with the task always completed and force rising to 93–101 N. Over-estimating it ($f \uparrow 1$) makes the robot conservative: harm falls because the task collapses (a no-op scores 0.000).

TABLE III
BOTH FAILURE DIRECTIONS, WHERE PREREGISTERED

axis	1/16	1/8	1/4	1/2	1	2	4	8	16
muscle strength	0.092	0.093	0.085	0.062	0.050	0.051	0.039	0.011	0.000
passive stiffness	0.090	0.092	0.084	0.064	0.050	0.044	0.034	0.016	0.001
reflex gain	0.062	0.062	0.062	0.063	0.050	0.027	0.043	0.008	0.029
task completion (muscle strength)	1.00	1.00	1.00	1.00	0.97	0.87	0.76	0.16	0.01

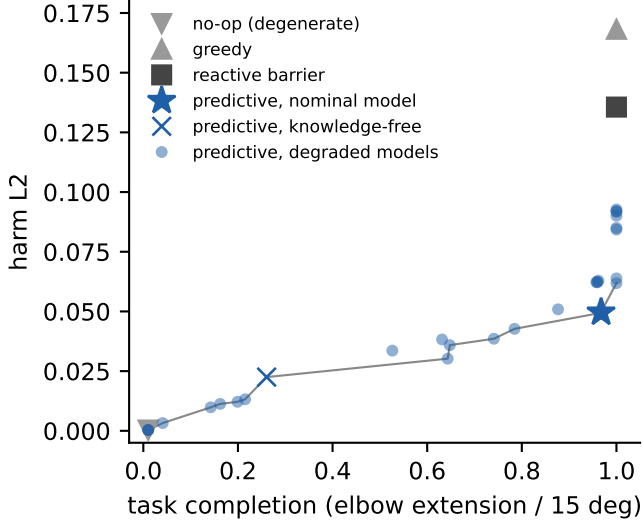


Fig. 2. Task completion vs harm for every arm and condition ($n = 19$), Pareto front drawn; no-op labelled as the degenerate solution.

TABLE IV
FORCE PREDICTION IS WHAT HOLDS THE LINE

condition	harm L2	peak force	paired vs constraint (0.136, 123 N)
nominal	0.049	50 N	-0.087 [-0.095, -0.079]
muscle strength 1/8	0.124	116 N	-0.011 [-0.018, -0.005]
passive stiffness 1/8	0.124	117 N	-0.012 [-0.020, -0.004]
knowledge-free	0.102	112 N (21% over ISO)	-0.034 [-0.062, -0.007]

Joint damping, segment mass and reflex delay move harm by at most 0.018 across the whole range; range of motion is one-sided (under-estimates paralyse the robot, over-estimates are invisible because the task never reaches the joint limit). Every paired difference against the constraint arm has its interval below zero: with both budgets active there is no harm crossover in [1/16, 16], the preregistered domain extension included.

4) Force prediction is what holds the line:

Remove the predictive arm's force budget (reflex budget only; constraint arm unchanged), 19 individuals (`crossover6_ext_n20noforce.json`):

Without force prediction the over-confident branch climbs to within 8% of the barrier. On the one-axis pilot (no force budget, planner with a binding reflex budget, 7 individuals) it crosses: at a four-fold reflex-gain under-estimate the predictive arm equals the greedy press (0.177, 214 N) and is more harmful than the constraint arm, +0.009 [+0.002, +0.015]. With force prediction the same six-degree-of-freedom rows stay at 0.084–0.093, a gap of 0.04–0.05, at every common limit from 80 to 160 N. The mechanism is physical: under

a stiff position servo the contact force is set mostly by the robot's own stiffness and the contact geometry, which the robot knows, so a copy whose muscles are wrong by sixteen-fold still predicts force well enough to stop.

5) *A freedom the wrong model misuses:* Letting the planner also choose the capsule orientation (along the bone or across it) helps no correct model and hurts every wrong one: copies that under-estimate the body choose "across" in 7 of 7 individuals and reach 0.11–0.12 at 109–122 N, above their own force budget by 10–22%, against 0.086–0.092 with the orientation fixed. Even the nominal copy chooses "across" in 3 of 7 and is worse for it (0.057 vs 0.050). An extra degree of freedom is an extra place to be wrong.

6) *A learned model in the same loop (one axis):* A network trained on observable quantities alone (16 inputs, next state and L2 as outputs, training speeds 0.02–0.10 m/s), dropped into the same planner at 0.25 m/s, is gentler than the barrier (paired -0.023 [-0.030, -0.015]) and less gentle than the physics copy (+0.016 [+0.006, +0.025]): about 60% of the copy's benefit while running outside its training speeds. An untrained network predicts a flat response, never binds and reproduces the greedy press exactly; a knowledge-free learned model is no better than no model, not worse. Clamping rollouts to the training box makes the network more aggressive, not less.

E. Limitations

- 1) **The proxy has no human validation.** Contact-evoked activation change is a person-specific protective response whose mapping to felt discomfort varies [15]; simulated rewards and human judgement have come apart before [29]. Rating material and the analysis are prepared (`out/render6/`); the study is pending ethics approval. Scope is neurologically intact or mildly impaired recipients; under complete spinal cord injury, spinal shock or sedation the reflex reads out neurological state, not contact noxiousness [19].
- 2) **The end-effector is idealised** (position-servoed joint chain, no manipulator); the capsule is symmetric about its axis so five of six degrees of freedom are outcome-relevant.
- 3) **No soft tissue;** pressure is a rigid-contact estimate.
- 4) **The internal model is a perturbed simulator.** This isolates parameter error with physical units, which is what turns a crossover into a specification, but does not exercise the failure modes of a deployed learned model beyond Section 4.6.
- 5) **Simulated gentleness is not a safety guarantee.** The deliverable is a specification, not a certificate.

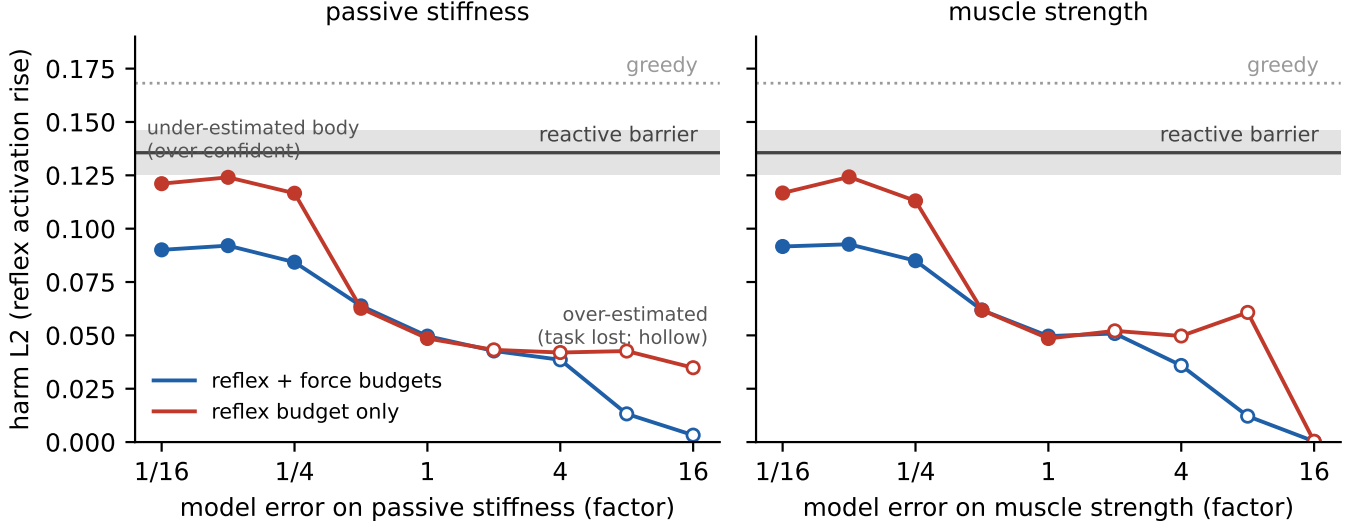


Fig. 1. Harm L2 against model error (log2, 1/16 to 16) for passive stiffness and muscle strength; horizontal band = reactive barrier ($n = 19$, paired-t band); dotted = greedy. Blue: both budgets; red: reflex budget only. Filled markers: task completion ≥ 0.95 ; hollow: task lost.

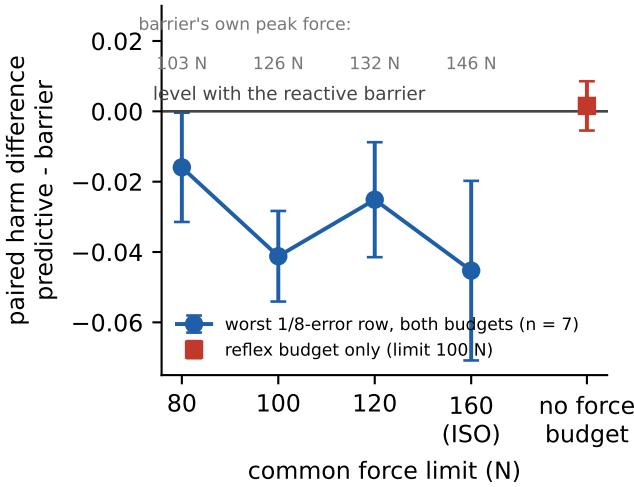


Fig. 3. Paired harm difference of the worst eight-fold under-estimation row against the barrier, as the common force limit is swept over 80/100/120/160 N ($n = 7$), with the barrier's own peak force annotated; the red point is the reflex-only ablation.

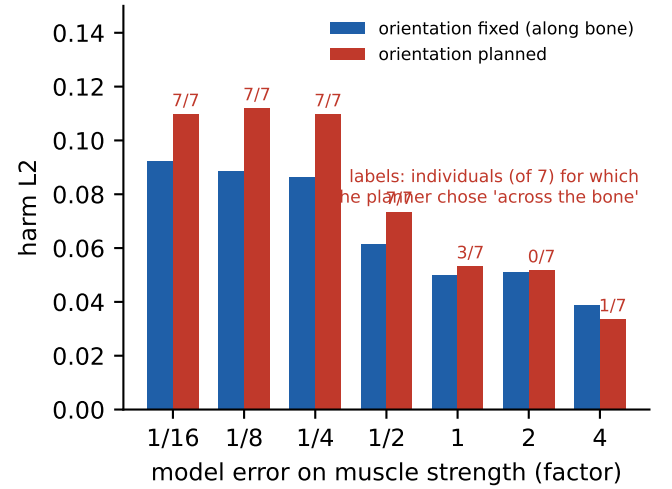


Fig. 5. Harm with the capsule orientation fixed along the bone versus planned, per muscle-strength error ($n = 7$); labels give how many individuals the planner chose "across" for.

F. Conclusion

A reactive force barrier is late; a predictive controller can be early but only as good as its model. Measured against an involuntary reflex on the recipient's side, the model needs to be right about two things, muscle strength and passive stiffness, to within about four-fold, and the robot must predict contact force as well as the response. Get those, and a sixteen-fold error on anything else does not make the touch worse than reacting would. Miss the force, and an eight-fold error brings prediction back to where reaction already is.

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